



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
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Ds Final Project

Digital System lab EE1501 2025

EE24BTECH11044 - MUTHYALA KOUSHIK
EE24BTECH11031 - JASHWANTH MEDAMONI
EE24BTECH11004 - ANKIT JAINAR

Contents

1	Problem Statement 1	3
2	Problem Statement - 2	5

1 Problem Statement 1

Design Specifications

Clock and Reset

- Base clock: 1s with 1ms precision
- Active-high reset initializes the system to default date/time

State Machine Design

- FSM with clearly defined states for clock display, date setting, time setting, timer, and alarm configuration
- State encoding uses 4-bit parameters
- Present and next state variables govern transitions

Time and Date Logic

- Time increments each second using the system clock
- Date update logic considers varying month lengths and leap years
- Time adjustment supported via increment/decrement inputs

12/24-Hour Mode

- Controlled by an input signal `mode_12_24`
- AM and PM outputs indicate period in 12-hour format

Timer and Alarm

- Timer supports adjustable countdown (0 to 600 seconds)
- Two buzzers: one for timer, another for alarm
- Alarm triggered at exact hour and minute

FSM State Transitions

Current State	Input Condition	Next State	Action/Description
CLOCK_DISPLAY	mode.button = 1	DATE_DISPLAY	Switch to date display
CLOCK_DISPLAY	set.button = 1	SET_TIME_HH	Begin time setting
CLOCK_DISPLAY	alarm.set = 1	SET_ALARM_HH	Begin alarm setting
CLOCK_DISPLAY	timer.start = 1	TIMER_SET	Begin timer setup
CLOCK_DISPLAY	else	CLOCK_DISPLAY	Remain in clock display

DATE_DISPLAY	mode_button = 1	CLOCK_DISPLAY	Return to clock display
DATE_DISPLAY	set_button = 1	SET_DATE_DD	Begin date setting
DATE_DISPLAY	else	DATE_DISPLAY	Remain in date display
SET_TIME_HH	set_button = 1	SET_TIME_MM	Move to setting minutes
SET_TIME_MM	set_button = 1	SET_TIME_SS	Move to setting seconds
SET_TIME_SS	set_button = 1	CLOCK_DISPLAY	Done setting time
SET_DATE_DD	set_button = 1	SET_DATE_MM	Move to setting month
SET_DATE_MM	set_button = 1	SET_DATE_YY	Move to setting year
SET_DATE_YY	set_button = 1	DATE_DISPLAY	Done setting date
SET_ALARM_HH	set_button = 1	SET_ALARM_MM	Move to setting alarm minutes
SET_ALARM_MM	set_button = 1	ALARM_ARMED	Finish setting alarm
TIMER_SET	timer_start = 1	TIMER_ACTIVE	Start countdown
TIMER_ACTIVE	timer_val == 0	CLOCK_DISPLAY	Timer done
ALARM_ARMED	alarm_buzzer && set_button = 1	CLOCK_DISPLAY	Acknowledge and stop alarm
Any state	reset = 1	CLOCK_DISPLAY	Reset system

Verilog Code Snippets

Refer to the code implementation provided in the appendix for full details. Example of state update block:

Listing 1: FSM State Update Block

```
always @(posedge clk or posedge reset) begin
    if (reset) present_state <= CLOCK_DISPLAY;
    else present_state <= next_state;
end
```

Testbench Strategy

- Simulate each feature: time increment, date roll-over, timer countdown, alarm trigger
- Use appropriate delays and signal triggers to demonstrate all states
- Observe buzzer activations and display changes

Conclusion

The digital clock design successfully integrates timekeeping, date tracking, alarm, and countdown functionalities using FSM principles in Verilog. It maintains modularity and clarity while offering options like 12/24-hour mode switching and system reset.

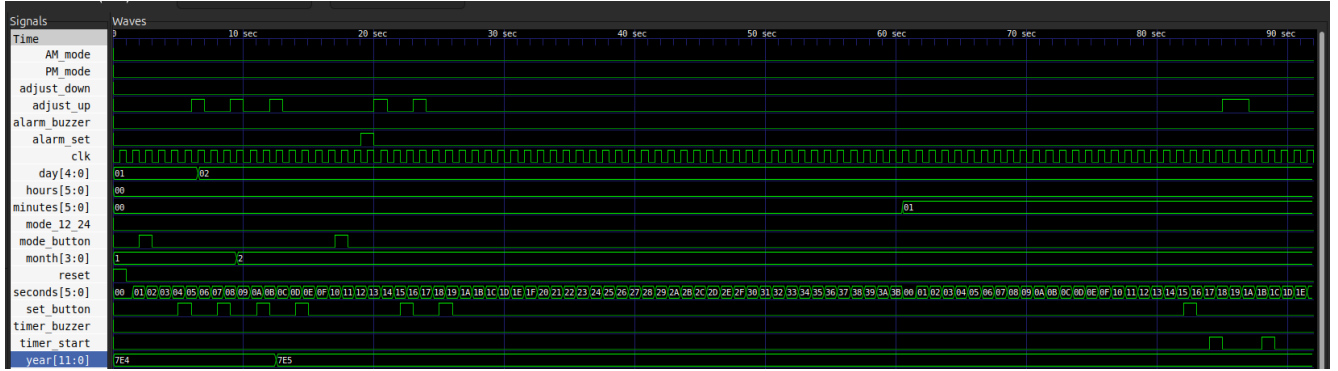


Figure 1: GTK-wave

2 Problem Statement - 2

The aim of this project is to design and test a Verilog-based traffic light controller system that simulates real-world traffic scenarios including normal operation, emergency vehicle handling, pedestrian walking signals, and alert mechanisms.

Functional Description

The traffic light system controls two roads: **T1** and **T2**, and provides:

- Standard Red-Yellow-Green traffic light control for each direction.
- Pedestrian walk signals (**T1_WALK** and **T2_WALK**).
- Buzzer alarm to indicate emergency vehicle clearance.
- Emergency override inputs (**emergency_left** and **emergency_right**) to prioritize ambulance or fire vehicles.
- Reset signal to initialize the system to a known state.

State Encoding

State Name	Binary Encoding	Decimal Encoding
STATE_IDLE	000	0
T1_GREEN_T2_RED	001	1
T1_ORANGE_T2_RED	010	2
T1_RED_T2_GREEN	011	3
T1_RED_T2_ORANGE	100	4
STATE_EMERGENCY_LEFT	101	5
STATE_EMERGENCY_RIGHT	110	6

Table 2: State Encoding for Traffic Light Controller

State Transition Table

Current State	Emergency Left	Emergency Right	Next State
STATE.IDLE	1	0	STATE.EMERGENCY_LEFT
STATE.IDLE	0	1	STATE.EMERGENCY_RIGHT
STATE.IDLE	0	0	T1.GREEN.T2.RED
T1.GREEN.T2.RED	0	0	T1.ORANGE.T2.RED
T1.GREEN.T2.RED	1	0	STATE.EMERGENCY_LEFT
T1.GREEN.T2.RED	0	1	STATE.EMERGENCY_RIGHT
T1.ORANGE.T2.RED	0	0	T1.RED.T2.GREEN
T1.ORANGE.T2.RED	1	0	STATE.EMERGENCY_LEFT
T1.ORANGE.T2.RED	0	1	STATE.EMERGENCY_RIGHT
T1.RED.T2.GREEN	0	0	T1.RED.T2.ORANGE
T1.RED.T2.GREEN	1	0	STATE.EMERGENCY_LEFT
T1.RED.T2.GREEN	0	1	STATE.EMERGENCY_RIGHT
T1.RED.T2.ORANGE	0	0	T1.GREEN.T2.RED
T1.RED.T2.ORANGE	1	0	STATE.EMERGENCY_LEFT
T1.RED.T2.ORANGE	0	1	STATE.EMERGENCY_RIGHT
STATE.EMERGENCY_LEFT	X	X	T1.RED.T2.GREEN
STATE.EMERGENCY_RIGHT	X	X	T1.GREEN.T2.RED

Table 3: State Transition Table for Traffic Light Controller

Key Features and Facilities

- **Safe Traffic Switching:** Ensures no conflicting green signals, using synchronized timing logic.
- **Emergency Handling:** Upon receiving an emergency input, the system overrides current state and grants green to the emergency direction, followed by controlled resume.
- **Pedestrian Safety:** Dedicated pedestrian walk signals with timing synchronized to traffic red phases.
- **Audio Alert (Buzzer):** Alerts when the system is in emergency mode, aiding pedestrians and drivers in decision-making.
- **Modular Design:** Clean Verilog module with testbench support for simulation and waveform verification.

Simulation and Verification

The design was tested using a Verilog testbench simulating:

1. Normal 100-second traffic cycle.
2. Emergency left and right triggers.
3. System behavior after emergencies.
4. Waveform verification using GTKWave with dumped VCD file.

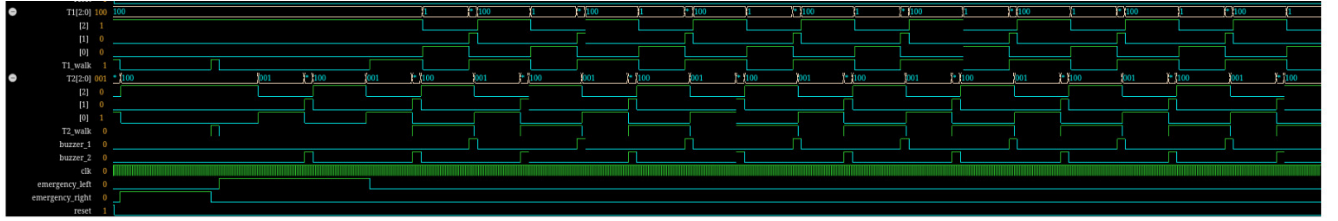


Figure 2: GTK-wave

Conclusion

The implemented traffic light controller successfully models an intersection with emergency vehicle prioritization and pedestrian safety measures. It is suitable for smart city applications and can be extended with traffic sensors or adaptive timing logic.